SAMPLING DISTRIBUTIONS



Learning outcomes

- Understand that sample statistics have a sampling distribution
- Perform point and interval estimation
- Interpret confidence intervals

Recap

We just explored properties of different probability models .

Example: We assumed that blood pressure is normally distributed with μ =120 and σ=10mm Hg.

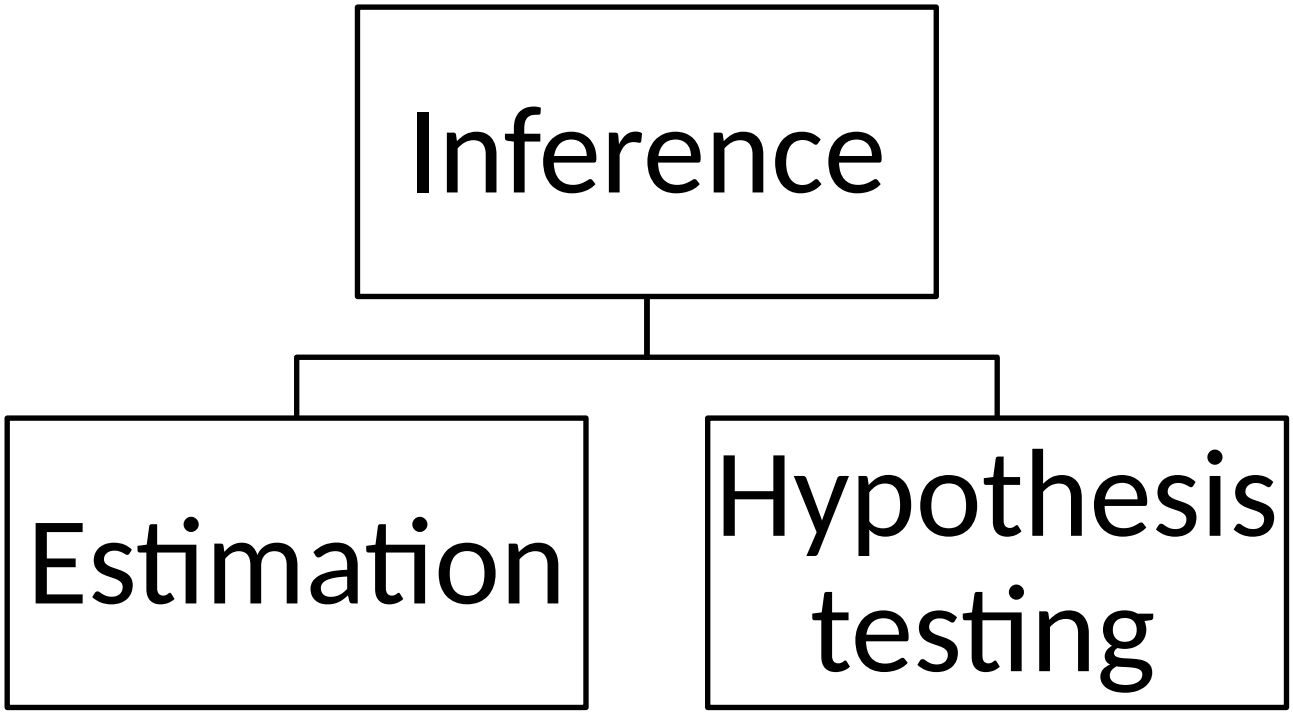
It was always assumed these properties were known and the only question left was what we can predict about the behaviour of the data given these properties.

What is the area of the curve between 100 and 140mm HG?

z_1= (100-120)/10=-2.00 and z_2=(140-120)/10 = 2.00. The area between -2 and +2 is .954. So 95.4% have a systolic BP between -2 and +2 standard deviations (between 100 and 140 mm Hg.

In practice…

- Have a dataset and want to *infer* properties of the underlying distribution from the dataset



Estimation

Example: Hypertension

Suppose we measure the systolic blood pressure (SBP) of a group of men patients attending a rural clinic and we believe the underlying distribution is normal. How can the parameters of this distribution (μ, σ^2) be estimated? How precise are our estimates?

Example: Infectious Disease

Suppose we look at people living within a low-income census tract in an urban area and we wish to estimate the prevalence of human immunodeficiency virus (HIV) in the community. We assume the number of cases among *n* people sampled is binomially distributed, with some parameter *p.* How is the parameter *p* estimated? How precise is this estimate?

Population and samples

Suppose we want to characterize the incubation period of a Salmonella outbreak in a population of infected individuals. Let the underlying distribution of incubation periods have a population mean μ and population variance σ^2.

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Ideally, we would like to estimate μ and σ^2 exactly using data from all individuals infected during the outbreak.

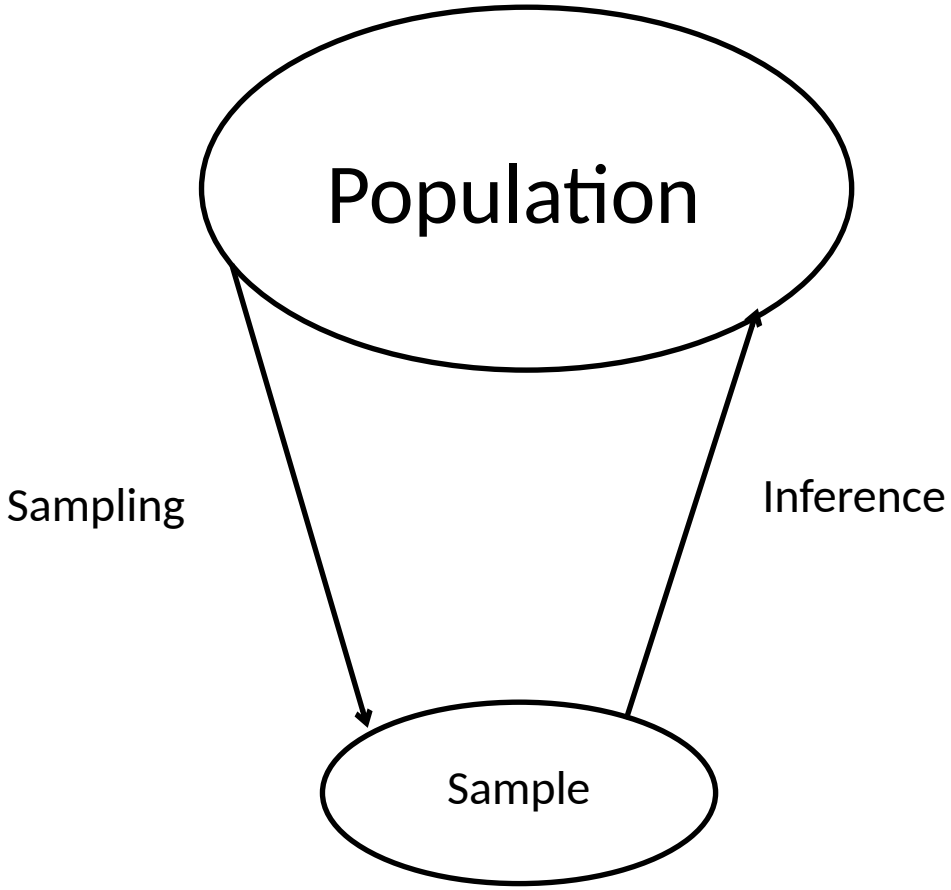
However, obtaining information from every infected individual may be impractical. Instead, we select a random sample of n infected individuals who are representative of the outbreak population and use their observed incubation periods,

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However, because studying every infected person is rarely feasible, we select a random sample of n individuals that is representative of the population and use their observed incubation periods,

$x_1, x_2, \ldots, x_n$ from this sample to help us estimate μ and σ^2.

Population and Samples



Population: 1,000 individuals infected during a Salmonella outbreak KwaZulu-Natal, 2024 Variable: Incubation period (days)

	2.9		3.1
2.7	2.76		3.15
3.7	3.3		3
3.2	4.1		3.2
3	2.9		3.5
2.65	3.16		3.3
3.5	2.7		2.95
3	3.1		2.8
2.194		2	
3.2	2.25		1.6
3	3.2		3.2
3	3.2		2.5
1.85	3		2.9
3.8	2.7		3.1
3.02	3		3.2
3.1	3		2.6
2.2	3		3
2.67	3.2		3.26
3.22	3.3		2.4
3.2	3.16		2.95
3.2	3.25		2.04
3.42	3.1		3.42
3.1	3.1		2.65
3.9	3.56		3
2.9	3.5		3.5
3.7	3.28		3.75
3	3.25		2.5
3.2	3.5		2.9
4.28	3		2.9
3.62	3.6		3.3
3.15			3
3.6	2.68		3.8
3.66	3.4		3.8
2.5	3.2		3.4
3	2.82		2.8
3.8	2.2		3.2
3.12	3.2		2.6
3.5	3.5		2.4
2.78	3.46		2.96
3.001		3.75	
1.7	2.42		2.9
2.65	3.1		2.1
2.7	3.38		3.1
2.8	3.25		3.9
2.1	3.9		3.3
3.6	2.9		2.75
2.45	3.3		3
3.5	3.9		3.85
3.1	3		2.14
3.1	1.001		3.48
3	2.9		3.96
3.15	3		3.1
3.42	3.22		2.7
3.9	3.7		2.82

Random samples of size 10

Individual	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
1	3	3.3	2.7	3.3	3.55
2	2.1	3.1	3.2	3.45	3
3	2	3.02	2.9	3.36	2.9
4	3.2	3.2	2.4	3.36	3
5	2.82	2.9	3.25	2.8	3.8
6	3.6	2.36	4.55	1.6	3.7
7	2.96	2.82	3.2	2.65	3.56
8	2.9	3	2.9	3.3	2.67
9	2.9	3.2	2.55	3.074	3.24
10	3.4	3.32	3.35	3.16	3.2
Mean	**2.831111**	**3.03**	**3.097**	**3.0244**	**3.258**
Standard deviation	**0.50091**	**0.295033**	**0.598685**	**0.566296**	**0.377501**

Estimation of the mean of the distribution

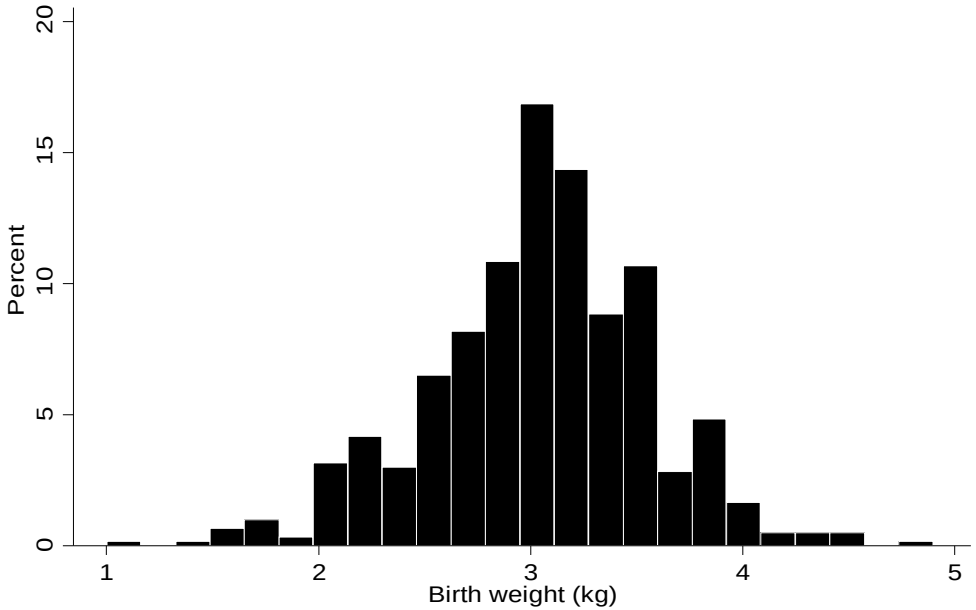
- The sample mean $\bar{X}$ is a random variable!
- So it has its own distribution
- Its own mean and standard deviation

Sampling distribution of the mean

- The sampling distribution of $\bar{X}$ is the distribution of values of $\bar{x}$ over all possible samples of size *n* that could have been selected from the reference population.

Back to the incubation period data

- Population distribution, mean 3.04 and standard deviation 0.52

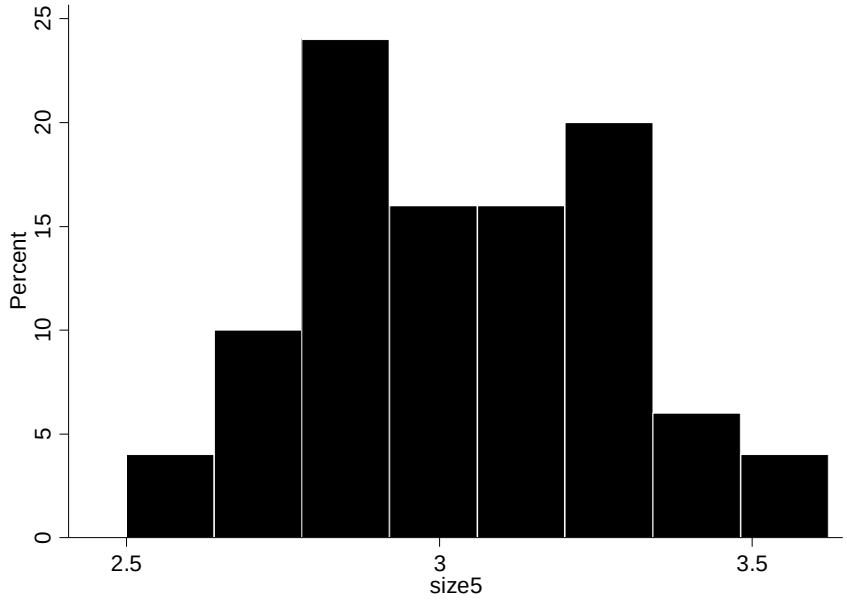


Sampling distribution of the me**an**

- 50 random samples of size 5 are drawn
- We compute the mean of each sample
- So now we have 50 means
- I can plot these and work out the means and standard deviation

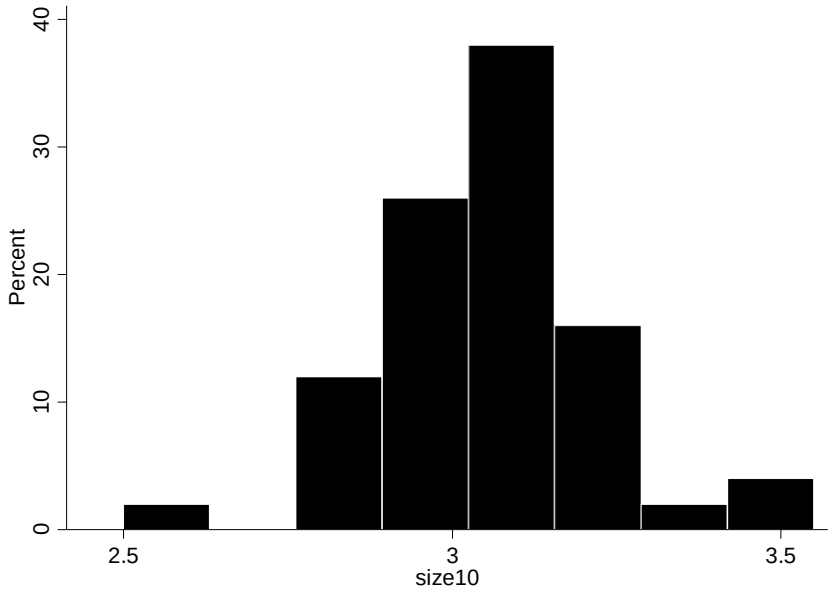
2.575	2.894	3.31
3.036	3.19	2.75
3.204	3.016	2.918
2.856	3.07	3.26
2.904	3.242	3.622
3.29	3.028	2.724
3.364	2.63	3.3
2.6848	2.97	2.786
3.122	2.66	
3.394	2.712	
2.82	3.182	
3.004	3.53	
3.028	3.4	
2.92	3.132	
3.212	3.29	
3.322	3.146	
2.82	2.884	
3.004	3.09	
2.89	3.036	
3.208	2.792	
3.17	2.81	

Means of 50 samples of size 10



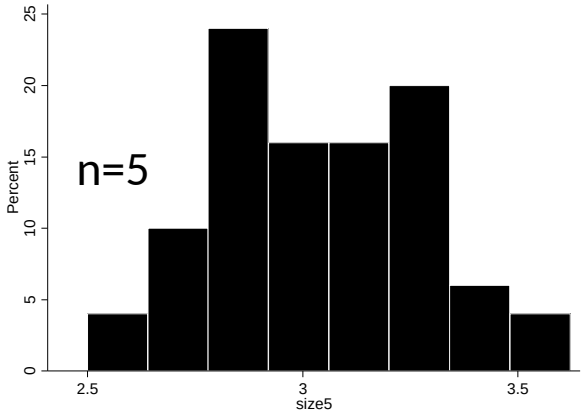
mean	3.044036 3
sd	0.244388

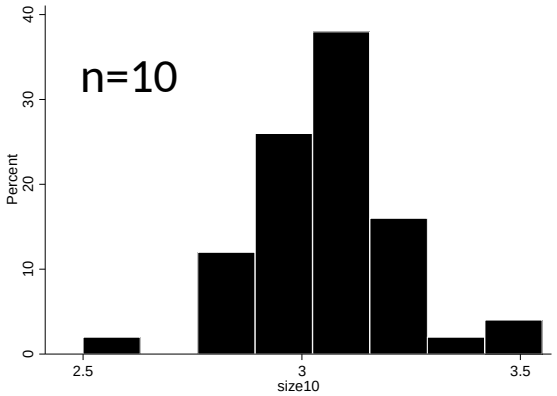
Means of 50 samples of size 10

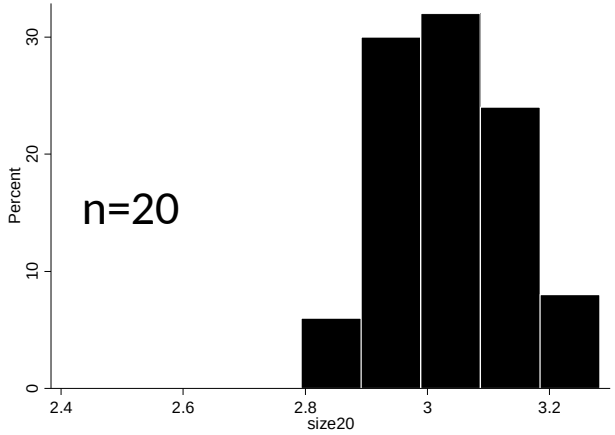


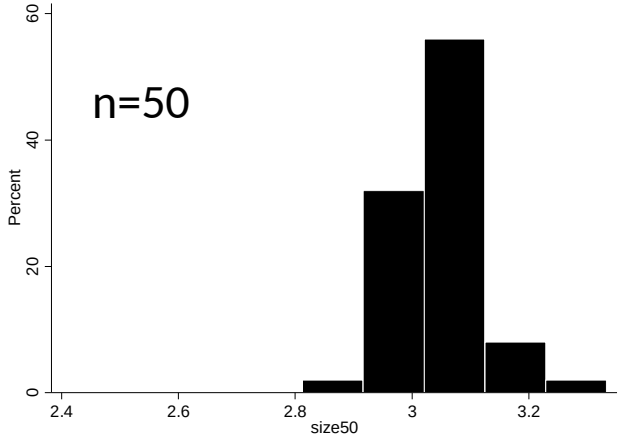
mean	3.049182
sd	0.170786

What do you notice?









stats	size5	size10	size20	size50
mean	3.044036	3.049182	3.038459	3.045317
sd	0.244388	0.170786	0.105266	0.07601

Properties of the sample mean

Let $X_1, \ldots, X_n$ be a random sample drawn from some population with mean μ. Then for the sample mean $\bar{X}$

$$E(\bar{X}) = \mu.$$

If $\bar{X}$ is the best estimator of μ then why do we prefer to estimate parameters from large samples instead of small ones?

Standard error of the mean

The larger the sample size the more precise the estimator is

$$Var(\bar{X}) = Var\left(\frac{\sum_{i=1}^{n} X_i}{n}\right) = \frac{1}{n^2}\sum_{i=1}^{n} Var\,(X_i)$$

But $Var\,(X_i) = \sigma^2$, therefore

$$Var(\bar{X}) = \left(\frac{1}{n^2}\right)(\sigma^2+\sigma^2+\ldots+\sigma^2) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

The set of sample means in repeated random samples of size *n* from this population has variance $\frac{\sigma^2}{n}$.

The standard deviation of this set of sample means is thus $\frac{\sigma}{\sqrt{n}}$ and is referred to as the *standard error of the mean* or the *standard error.*

Estimation of the standard error of the mean

In practice, the population variance σ^2 is rarely known. A reasonable estimator for the population variance σ^2 is the sample variance s^2, which leads to the following definition:

The standard deviation of this set of sample means is $\frac{\sigma}{\sqrt{n}}$ and is estimated by $\frac{s}{\sqrt{n}}$.

The Central Limit Theorem

- If the underlying distribution is normal, then it can be shown that the sample mean is itself normally distributed with mean μ and variance $\frac{\sigma^2}{n}$
- In other words, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

What if the underlying distribution is not normal?

Central-Limit Theorem

Let $X_1, \ldots, X_n$ be a random sample from some population with mean μ and variance σ^2.

Then for large *n*, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ even if the underlying distribution of individual observations in the population is not normal.

(The symbol ~ is used to represent "approximately distributed.")

Application of the central limit theorem

- If $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ then we can re-express it in standardized form by

$$Z = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}}$$

Example: Central Limit Theorem

Compute the probability that the mean incubation period from a random sample of 10 individuals infected during the Salmonella outbreak falls between 2.72 and 3.36 days (i.e. $2.72 \leq \bar{X} < 3.36$) sssuming that the population of 1,000 infected individuals has a mean incubation period of 3.04 days and a standard deviation of 0.52 days.

If the central limit theorem is applied we assume $\bar{X}$ follows a normal distribution with mean μ=3.04 and standard deviation $\frac{\sigma}{\sqrt{n}}=\frac{0.52}{\sqrt{10}}=0.16$

$P(3.36 \leq \bar{X} < 2.72) = P(\bar{X} < 3.36) - P(\bar{X} < 2.72)$

$= P(Z < \frac{3.36-3.04}{0.16}) - P(Z < \frac{2.72-3.04}{0.16})$

$= P(Z < 2) - P(Z < -2)$

$= 0.97724987 - 0.2275013$

$= 0.954$

Interval Estimation

Recap:

- We discussed using $\bar{x}$ to estimate μ
- Variability of the sample mean - standard error

However, we frequently wish to obtain an interval of plausible estimates of the mean as well as a best estimate of its precise value.

Interval Estimation

Individual	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
1	3	3.3	2.7	3.3	3.55
2	2.1	3.1	3.2	3.45	3
3	2	3.02	2.9	3.36	2.9
4	3.2	3.2	2.4	3.36	3
5	2.82	2.9	3.25	2.8	3.8
6	3.6	2.36	4.55	1.6	3.7
7	2.96	2.82	3.2	2.65	3.56
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Standard deviation	**0.50091**	**0.295033**	**0.598685**	**0.566296**	**0.377501**

Suppose the first sample of 10 incubation periods given above has been drawn. Our best estimate of the population mean would be the sample mean 2.83. Although 2.83 days is our best estimate of μ we still are not certain that μ is 2.8 days. Indeed, if the second sample of 10 birth weights had been drawn, a point estimate of 3.03 would have been obtained. Our point estimate would certainly have a different meaning if it was highly likely that μ was within 1 day instead of 0.10 day of the 2.83 days

Two Sided Confidence Intervals

- For a standard normal distribution 95% of observations fall between -1.96 and 1.96. So,

$$P(-1.96 \leq Z \leq 1.96) = 0.95$$

But we know that $Z = \dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}}$

$$P(-1.96 \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq 1.96) = 0.95$$

$$P(-1.96\ \sigma/\sqrt{n} \leq \bar{X} - \mu \leq 1.96\sigma/\sqrt{n}) = 0.95$$

$$P(-1.96\frac{\sigma}{\sqrt{n}} - \bar{X} \leq -\mu \leq 1.96\sigma/\sqrt{n} - \bar{X}) = 0.95$$

Two Sided Confidence Intervals

$$P(1.96\frac{\sigma}{\sqrt{n}} + \bar{X} \geq \mu \geq -1.96\sigma/\sqrt{n} + \bar{X}) = 0.95$$

Rearranging the terms:

$$P(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96\sigma/\sqrt{n}) = 0.95$$

Now the probability statement says something about μ. The quantities $\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}$ and $\bar{X} + 1.96\frac{\sigma}{\sqrt{n}}$ are the 95% confidence limits for the population mean.

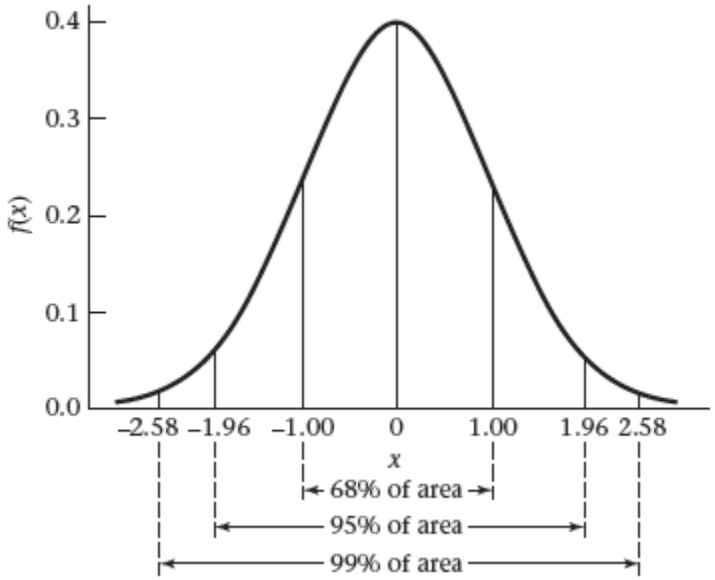
Two sided confidence intervals

So, we are 95% confident that the interval $(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96\frac{\sigma}{\sqrt{n}})$ will cover μ.

This means that if we were to select 100 different samples from the population and use these samples to calculate 100 confidence intervals, approximately 95 of these will cover the true population mean and 5 will not.

What if we want to be more certain than 95%

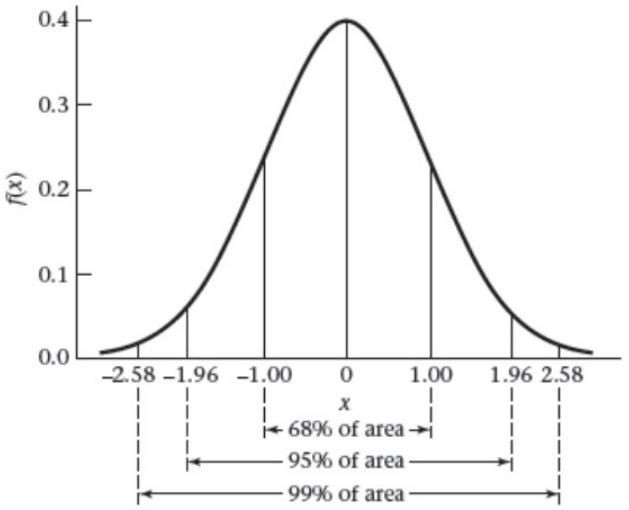
- Although 95% CIs is most commonly used in practice, we are not restricted to this choice
- We might prefer a greater degree of certainty
- We could choose a 99% CI instead



What if we want to be more certain than 95%

- Since 99% of observations lie between -2.58 and 2.58, a 99% CI for μ *is*

$$(\bar{X} - 2.58\frac{\sigma}{\sqrt{n}}, \bar{X} + 2.58\frac{\sigma}{\sqrt{n}})$$



Is this wider or narrower than the 95% CI?

Generic confidence interval for μ

$$(\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}})$$

This interval then has a $100\ (1-\alpha)\%$ of covering μ.

The t-distribution

- So far we have been assuming that σ was known
- This is rarely the case in practice
- When σ is unknown we need to substitute s (the standard deviation for the sample) for σ.

$$Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

~~$$Z=\frac{\bar{X}-\mu}{s/\sqrt{n}}$$~~

The t-distribution

If X is normally distributed and a sample of size n is chosen from this underlying population, the probability distribution of

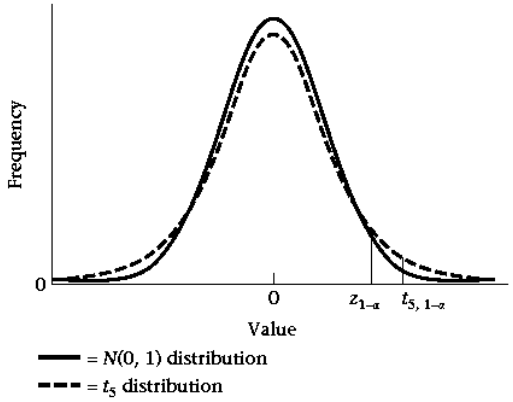
$$t=\frac{\bar{X}-\mu}{s/\sqrt{n}}$$

is known as Student's t distribution with $n-1$ degrees of freedom. We represent it as t_{n-1}.

Properties of the t distribution

- The *t* distribution is symmetric about 0 but is more spread out than the $N(0, 1)$ distribution
- The difference between the *t* distribution and the normal distribution is greatest for small values of *n* (*n* < 30).

Comparison of Student's *t* distribution with 5 degrees of freedom with an *N*(0, 1) distribution



Finding percentiles of the t distribution

Just like how we found the upper 5% of a standard normal distribution, we can do the same for the t-distribution.

Find the upper 5th percentile of a *t* distribution with 23 *degrees of freedom*.

Table 5 Percentage points of the t distribution ($t_{d,u}$)*

Degrees of freedom, d	u								
	.75	.80	.85	.90	.95	.975	.99	.995	.9995
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	31.598
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.767
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.551
60	0.679	0.848	1.046	1.296	1.671	2.000	2.390	2.660	3.460
120	0.677	0.845	1.041	1.289	1.658	1.980	2.358	2.617	3.373
∞	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.291

*The uth percentile of a t distribution with d degrees of freedom.

Source: Table 5 is taken from Table III of Fisher and Yates: "Statistical Tables for Biological, Agricultural and Medical Research," published by Longman Group Ltd., London (previously published by Oliver and Boyd Ltd., Edinburgh). Reprinted by permission of Pearson Education Ltd.

A 100% × (1 − α) CI for the mean μ of a normal distribution with unknown variance

$$(\bar{X} - t_{n-1;1-\alpha/2}\frac{s}{\sqrt{n}}, \bar{X} + t_{n-1;1-\alpha/2}\frac{s}{\sqrt{n}})$$

In shorthand notation :

$$\bar{X} \pm t_{n-1;1-\alpha/2}\frac{s}{\sqrt{n}}$$

Factors Influencing the Length of a Confidence Interval

$$\text{Length} = 2t_{n-1,\,1-\alpha/2}\frac{s}{\sqrt{n}}$$

1. α: level of confidence; as α gets smaller, CI gets longer
2. s: as $s \uparrow$ length $\uparrow$
3. n: as $n \uparrow$ length $\downarrow$

Example

- Back to the incubation period data, suppose that we selected Sample 1 and wish to construct a 95% CI for the population mean where σ is unknown.

Individual	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
1	3	3.3	2.7	3.3	3.55
2	2.1	3.1	3.2	3.45	3
3	2	3.02	2.9	3.36	2.9
4	3.2	3.2	2.4	3.36	3
5	2.82	2.9	3.25	2.8	3.8
6	3.6	2.36	4.55	1.6	3.7
7	2.96	2.82	3.2	2.65	3.56
8	2.9	3	2.9	3.3	2.67
9	2.9	3.2	2.55	3.074	3.24
10	3.4	3.32	3.35	3.16	3.2
Mean	**2.831111**	**3.03**	**3.097**	**3.0244**	**3.258**
Standard deviation	**0.50091**	**0.295033**	**0.598685**	**0.566296**	**0.377501**

Example (cont.)

$$(\bar{X} - t_{n-1;1-\alpha/2}\frac{s}{\sqrt{n}}, \bar{X} + t_{n-1;1-\alpha/2}\frac{s}{\sqrt{n}}) \qquad (2)$$

The sample of n=10 individuals has mean 2.83 and standard deviation 0.50. That is $\bar{x}$=2.83 and $s = 0.50$. From the t-table we have

$t_{9;0.975}$ =2.262. Substituting into (2)

$$(2.83 - 2.262\frac{0.5}{\sqrt{10}}, 2.83 + 2.262\frac{0.5}{\sqrt{10}})$$

$$(2.83 - 0.3576, 2.83 + 0.3576)$$

$$(2.4724, 3.1876)$$

100% × (1 − α) Confidence interval for a proportion

- Recall that when $np > 5$ and $n(1-p) > 5$, the sampling distribution of $\hat{p}$ is approximately normal with mean p and standard deviation

$$\sigma_p = \sqrt{\frac{p\,(1-p)}{n}}$$

100% × (1 − α) Confidence interval for a proportion

$$(\hat{p} - z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}})$$

Or

$$\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Example

A surveillance study of tuberculosis (TB) contacts enrolled 591 household contacts of patients with active TB. Among these contacts, 204 tested positive for latent TB infection using an interferon-gamma release assay (IGRA). We wish to construct a 95% confidence interval for the proportion of household contacts with latent TB infection in the sampled population.

Example (cont.)

$$\hat{p} = \frac{204}{591} = 0.3452$$

Since

$n\hat{p} = 591 \times 0.3452 = 204 > 5$ and $n(1-\hat{p}) = 591\,(1 - 0.3452) = 387 > 5$

we can use the normal distribution

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.3452\,(1-0.3452)}{591}} = 0.0196$$

Substituting into $(\hat{p} - z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}},\ \hat{p} + z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}})$

$$\left(0.3452 - 1.96(0.0196), 0.3452 + 1.96(0.0196)\right)$$

$$= (0.3086, 0.3836)$$

Example (cont.)

We are 95% confident that the population proportion p of household contacts with latent TB infection belongs to the interval $(0.3068, 0.3836)$.

In other words, based on this sample, we estimate that between 30.68% and 38.36% of all household contacts in the target population have latent TB infection, and we are 95% confident that this interval contains the true population proportion.